



SAFER: System for Disaster Assessment and Forecasting of Emergency Risks using Geo-Spatial Data

Group No. 8

Sonal Sonarghare	21107033
Janhavi Kasar	21107030
Aman Dhumal	21107036
Vedant Mayekar	22207007

Project Guide
Prof. Anagha Aher

Contents

- Abstract
- Introduction
- Objectives
- Literature Review
- Research Gap
- Problem Definition
- Scope
- Technological Stack
- Proposed System Architecture/Working
- Implementation

Abstract

Disaster management in Northeast India faces challenges like frequent natural disasters, inadequate infrastructure, limited real-time data, and poor communication systems. Resource constraints and insufficient emergency services further complicate preparedness and response efforts in the region. SAFER (System for Disaster Assessment and Forecasting of Emergency Risks using Geo-Spatial Data) is a web and mobile platform designed to enhance disaster management in Northeast India, a region prone to floods and landslides. It uses satellite data (Sentinel 1 & 2) and an Attention-based LSTM model for accurate assessment and early prediction of disasters. Key features include a voice-controlled geospatial interface, real-time alerts, dynamic reports, and an intelligent ambulance dispatch system. SAFER supports SMS-based communication for low-connectivity areas, improving emergency response, coordination, and accessibility in remote regions. .

Introduction

1. Problem Domain Overview

- **High Vulnerability:** Northeast India's hilly terrain, fragile ecosystem, and heavy rainfall make it prone to natural disasters.
- **Severe Impact:** Frequent floods and landslides cause major damage to lives, infrastructure, and the local economy.
- **Recurrent Disasters:** Annual floods and landslides lead to displacement, property loss, and isolation of remote communities.



Introduction

2. Challenges in Current Disaster Management Systems

- **Lack of Real-Time Data:** Limited access to timely data hampers quick disaster prediction and response.
- **Fragmented Information:** Incomplete integration of geospatial data weakens disaster assessment accuracy.
- **Unreliable Communication:** Breakdowns in communication systems slow down coordination during emergencies.
- **Limited Resources:** Shortages in emergency services and medical assistance challenging.

Introduction

3. Current Solutions and Persisting Issues

- **Non-Intuitive Interfaces:** Complex systems make it difficult for non-expert users to operate in real-time scenarios.
- **Static and Incomplete Data:** Reliance on historical data and limited integration of real-time inputs hampers disaster prediction and tracking.
- **Lack of Real-time Information:** Inefficient distribution of updates delays crucial information to responders and the public.
- **Inadequate Ambulance Integration:** Many solutions lack seamless coordination between disaster forecasting and ambulance services, causing delays in medical aid.

Introduction

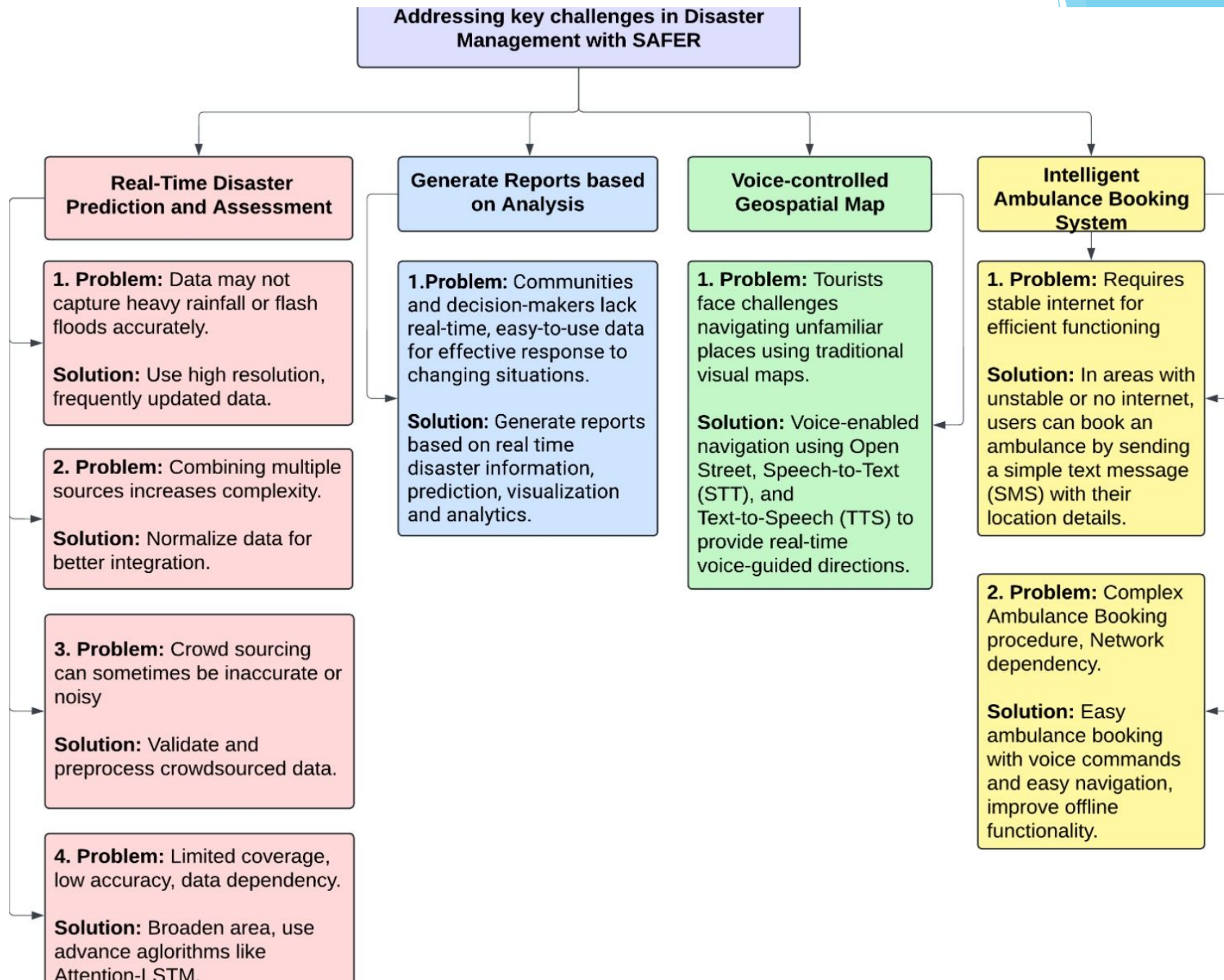
4. Motivation

- **Frequent Disasters:** Recurrent floods, landslides, and earthquakes in Northeast India highlight the need for improved disaster management.
- **Existing Limitations:** Current systems face delays in rescue operations due to poor communication, insufficient real-time data, and inefficient ambulance dispatch systems.
- **Complex Interfaces:** Existing solutions are difficult for non-expert users, especially in areas with low digital literacy.
- **Urgency for Improvement:** The need for real-time, user-friendly solutions to enhance emergency preparedness and reduce response times.

Objectives

1. To integrate disaster assessment and prediction in Northeast India using an Attention-based LSTM network, incorporating both raster and vector spatial data.
2. To generate comprehensive reports and statistics based on analysis.
3. To provide a voice-controlled interface for seamless interaction with geospatial maps using Web Speech API and Flutter STT.
4. To integrate an Intelligent Ambulance Booking System with real-time GPS tracking and a Voice-Controlled First Aid Assistant.

Literature Review



Literature Review

Sr.no	Title	Author(s)	Year	Methodology	Drawback
1	Satellite Flood Inundation Assessment and Forecast Using SMAP and Landsat	Jinyang Du , John S. Kimball, Senior Member, Justin Sheffield, Ming Pan Colby K. Fisher, Hylke E. Beck, and Eric F. Wood	2021	A flood prediction model using rainfall data, soil moisture data, and flood events observed via satellite data, with forecasts up to 3 days using a Classification and Regression Trees (CART) model.	Accuracy Depends on data resolution but increase in accuracy will need more computational power. Short-term events (heavy rainfall for 1hr or flash floods) may not be captured accurately if the data is not updated frequently enough.
2	Flood Forecasting System Based on Integrated Big and Crowdsourced Data by Using Machine Learning Techniques	Supattra Puttinaovarat, Paramate Horikawa	2020	Develop a flood forecasting system that combines real-time and crowdsourced data with machine learning techniques to deliver accurate and timely flood predictions.	Accuracy depends on data quality Combining multiple data sources increases complexity . Crowdsourcing can sometimes be inaccurate or noisy .

Literature Review

Sr.no	Title	Author(s)	Year	Methodology	Drawback
3	Geospatial techniques for flood inundation mapping	Kuldeep, P. K. Garg, R. D. Garg	2016	A framework to identify river islands suitable for eco-friendly tourism by analyzing satellite data and flood risks to ensure these sites are Resistant to flooding.	Accuracy of flood modeling and land classification depends on the resolution . HEC-RAS is Resource Demanding.
4	Landslide Detection Using Deep Learning and Object-Based Image Analysis	Omid Ghorbanzadeh, Hejar Shahabi, Alessandro Crivellari, Saeid Homayouni, Thomas Blaschke, and Pedram Ghamisi	2022	The paper proposes an integration framework combining Deep Learning (ResU-Net) and Object-Based Image Analysis (OBIA) for landslide detection using Sentinel-2 satellite imagery and topographic data. The ResU-Net model generates a heatmap of landslide probabilities, while OBIA refines the results by analyzing image objects instead of individual pixels, improving the precision and overall detection accuracy.	1. Pixel-based Deep Learning (DL) models can struggle to differentiate between similar-looking features (e.g., riverbeds vs. landslides). This limitation often leads to false positives in the landslide detection process.

Literature Review

Sr.no	Title	Author(s)	Year	Methodology	Drawback
5	Predicting and Understanding Landslide Events With Explainable AI	Enrico Collini, L. A. Ipsaro Palesi, Paolo Nesi, Gianni Pantaleo, Nicola Nocentini, Ascanio Rosi.	2022	The paper presents a 1-day landslide prediction model using XGBoost, incorporating static and real-time features like rainfall, temperature, humidity, and vegetation. Explainable AI (XAI) is applied to assess feature relevance both globally and locally. Data collection, processing, and publication are managed via the Snap4City infrastructure, with the model validated using data from 2013-2019 in the Metropolitan City of Florence.	1. While the model achieves better prediction accuracy than previous approaches, its implementation is computationally expensive due to the large dataset and the heterogeneity of the input data.
6	Landslide detection in Uttarakhand region using active remote sensing	V. Chamoli, R. Bahuguna, R. Gowri, R. Prakash, A. Vidyarthi and V. P. Dubey	2024	Uses Sentinel-1 data and ERDAS Imagine classification.	1. Limited to Tota Ghati. 2. Low Kappa score. 3. Sentinel-1 dependency.

Literature Review

Sr.no	Title	Author(s)	Year	Methodology	Drawback
7	Building community resilience through geospatial information dashboards	S. Praharaj, E. Wentz	2022	The methodology involved collecting relevant local data via APIs, integrating it using a four-layer architecture with GeoNode for storage and R for data processing. A multi-view web dashboard with interactive visualizations and data segregation options was developed to help communities understand and respond to issues effectively.	1. Lack of Granular Data: Dashboards often lack sub-group data, limiting insights into the unequal impacts of crises. 2. Complex Navigation: Poor design, limited data options, and unclear narratives reduce user engagement and decision-making effectiveness.
8	An Interactive Voice Assistant System for guiding the Tourists in Historical places	R. Keerthana, Dr. T. Ananth Kumar, P. Manjubala, M. Pavithra	2020	The methodology involves a voice assistant system that converts user voice commands into text, matches it with a database of location data, and provides real-time navigation using Text-to-Speech (TTS) technology, guiding users through audio instructions to their destination.	1. Internet and Language Limitations: The system's reliance on a stable internet connection and limited language support restricts its usability in areas with poor connectivity and for non-English speakers. 2. Accuracy and Privacy Concerns: Speech recognition may face challenges in noisy environments or with different accents, and the use of location data raises potential privacy issues if not securely managed.

Literature Review

Sr.no	Title	Author(s)	Year	Methodology	Drawback
9	Development of a Voice Virtual Assistant for the Geospatial Data Visualization Application on the Web	Homeyra Mahmoudi, Silvana Camboim, Maria Antonia Brovelli	2023	The authors built a voice-controlled interface for the BStreams platform using PlanGraph theory and user surveys to understand geospatial language. They validated commands with ChatGPT ($r = 0.81$) and implemented the system with JavaScript tools, followed by usability testing with 10 users.	1. Accent Sensitivity: Voice commands were often misinterpreted due to accent variations. 2. Domain Knowledge Dependency: Non-expert users struggled to give accurate voice commands. 3. Limited Multilingual Support: The system mainly supported English, reducing accessibility.
10	The Ambulance Booking and Tracking System	Anbarasu P Arunkumar S Gowtham R Deepak S Latha M (Professor)	2024	A mobile app allows users to book ambulances, which are tracked in real time via GPS. A server processes requests, matches ambulances, and sends updates. Notifications provide estimated arrival time, and a database stores relevant data.	1. Internet Dependency: Requires stable internet for efficient functioning. 2. Limited Coverage: Less effective in rural areas with poor network or ambulance availability.

Literature Review

Sr.no	Title	Author(s)	Year	Methodology	Drawback
11	Ambulance Booking System	R. Jyothsingh (Assistant Professor, SRK Institute of Technology) Thota Swathi Kareti Lepakshi Pulivarthi Akash Puvvaladasu Hemanth	2024	The Ambulance Booking System uses a mobile app where users enter emergency details to request an ambulance. The system processes these requests, matches them with available ambulances, and provides real-time tracking and notifications about the ambulance's status and arrival.	1. Internet Reliance: Depends on stable internet for functionality. 2. Limited Features: May lack advanced features compared to more comprehensive systems
12	Optimal Ambulance Positioning for Road Accidents with Deep Embedded Clustering	Dhyal Desai Dhyani Joyeeta Dey Sandeep Kumar Satapathy Shruti Mishra Sachi Nandan Mohanty Pallavi Mishra Sandeep Kumar Panda	2023	The study proposes using Deep Embedded Clustering to optimize ambulance positioning for quick response to road accidents. The system clusters accident-prone areas using historical data and positions ambulances accordingly, reducing response time.	1. Data Dependency: Requires extensive and accurate historical accident data.

Research Gap (Limitations of existing systems)

Current disaster assessment and forecasting frameworks are not fully implemented as apps or websites, limiting their interactivity.

They often lack precision, especially for short-duration events like intense rainfall, which can be improved by using higher-resolution data.

Techniques like the weighted-based approach oversimplify complex interactions in landslide prediction. Advanced machine learning techniques should be used to model these interactions more effectively.

Crowd-sourced data can be noisy and unreliable without proper validation.

Existing systems face delays in real-time information distribution, and there is a need to improve the integration of emergency services with real-time tracking systems.

Problem Definition

- **Inaccurate Disaster Assessment and Forecasting:** The existing methods for assessing and forecasting disasters lack precision due to inadequate integration of raster and vector spatial data and outdated predictive models.
- **Non-Intuitive User Interfaces:** Existing disaster management systems have complex, non-user-friendly interfaces, making it difficult for users to interact with geospatial data effectively.
- **Delayed Real-Time Information Distribution:** There is a lack of timely updates and real-time information dissemination, which hampers effective decision-making and response during emergencies.
- **Limited Emergency Services Integration:** Emergency services lack integration with real-time tracking, affecting the efficiency of overall emergency response.

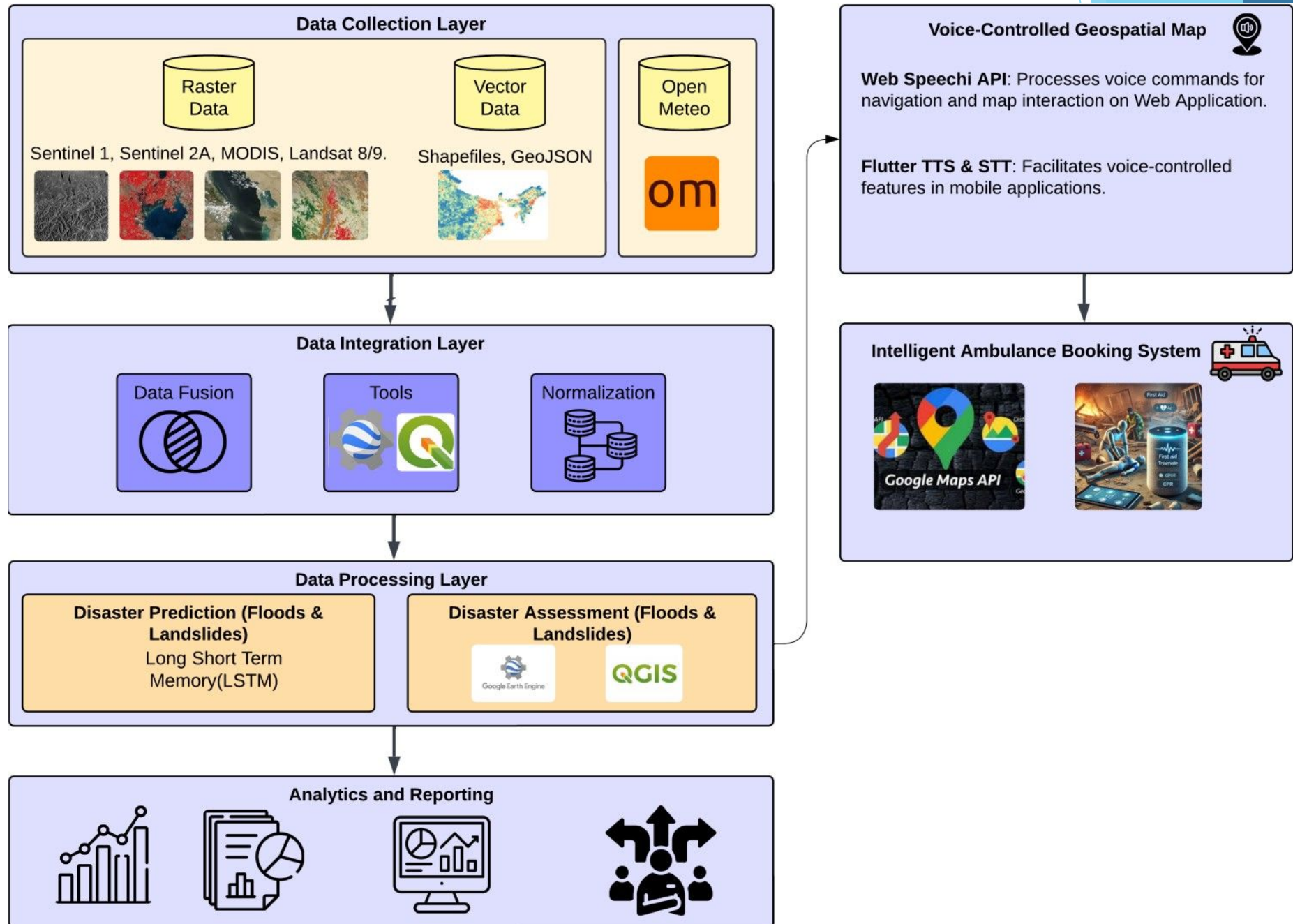
Scope

- Provide **Emergency Responders** with real-time disaster information and predictions.
- Assist **Local Governments** in planning and allocating resources for disaster preparedness.
- Enhance dispatch efficiency with smart ambulance booking and alert **Emergency Rescue teams**.
- Allow the **public** to access disaster forecasts and locate safe zone.
- Help **Disaster Management Authorities** make informed decisions with predictive models and analytics.

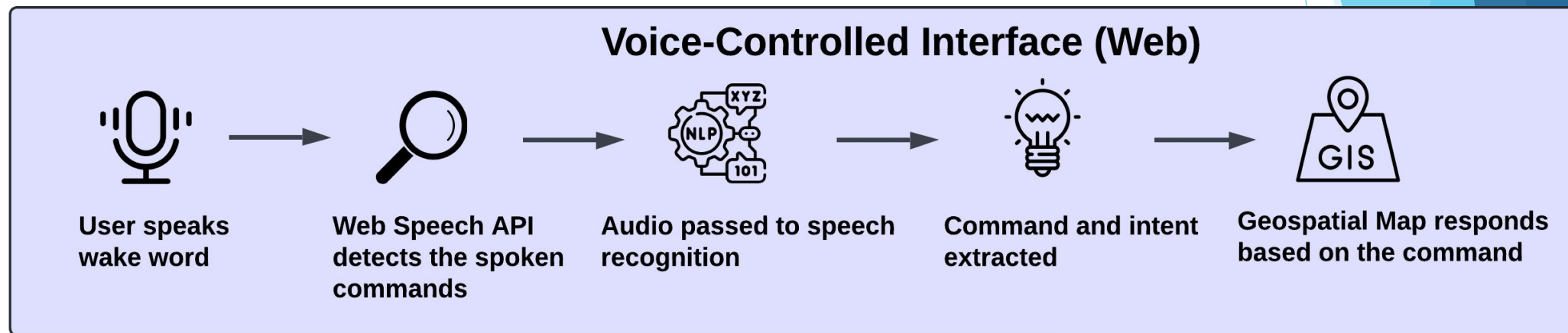
Technological Stack

- **Frontend:** Flutter 3.22 (App)
- **Framework:** Django 5.1.1 (Web)
- **Input Data:**
 - i. **Raster Data:** Sentinel 1, Sentinel 2A, Landsat 8/9
 - ii. **Vector Data:** Shapefiles, GeoJSON
 - iii. **Weather Data:** OpenMeteo
- **Disaster Assessment and Prediction:** Google Earth Engine, Attention-based LSTM model
- **Voice-Controlled Geospatial Map:** Web Speech API, Open street map
- **Intelligent Ambulance Booking:** Open street map

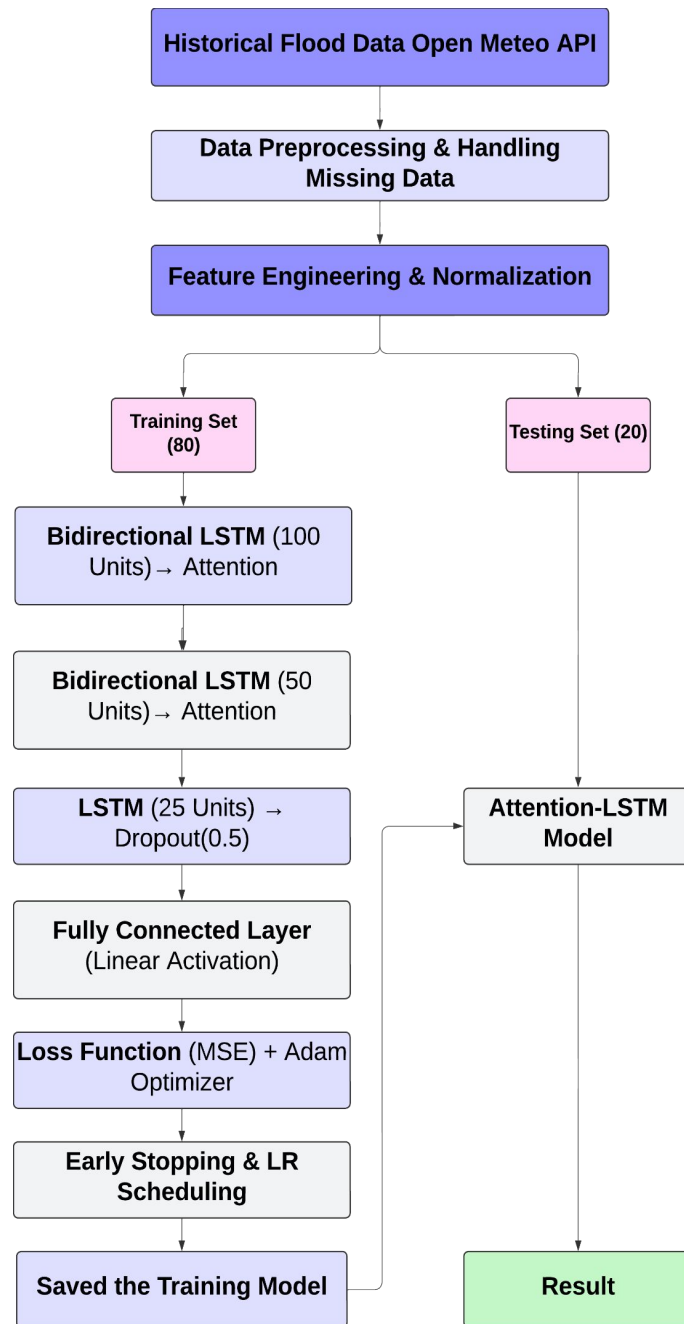
Proposed system architecture/Working



Proposed system architecture/Working



Proposed system architecture/Working



Implementation Status

SAFER

077

Ambernath Taluka

37.84% Flood Probability

Stay informed about flood risks in your area. Flooding can occur unexpectedly, especially during heavy rainfall or rapid snowmelt. It's crucial to know the signs of potential flooding and to be prepared.

Safety Tips: Don't drive through flooded roads. [more](#)

Emergency Services: 108
Local Flood Helpline: 1077

[Learn More](#)

AREAS OF SUPPORT

Hover over a card to explore



Ambulance

Need an ambulance at your current location? Click below for more information.

[Learn more](#)



Quick First Aid

Use our quick first aid assistant for immediate help until the ambulance arrives.

[Learn more](#)



Disaster Assessment

Access our disaster assessment tool for critical information on post-disaster impacts.

[Learn more](#)

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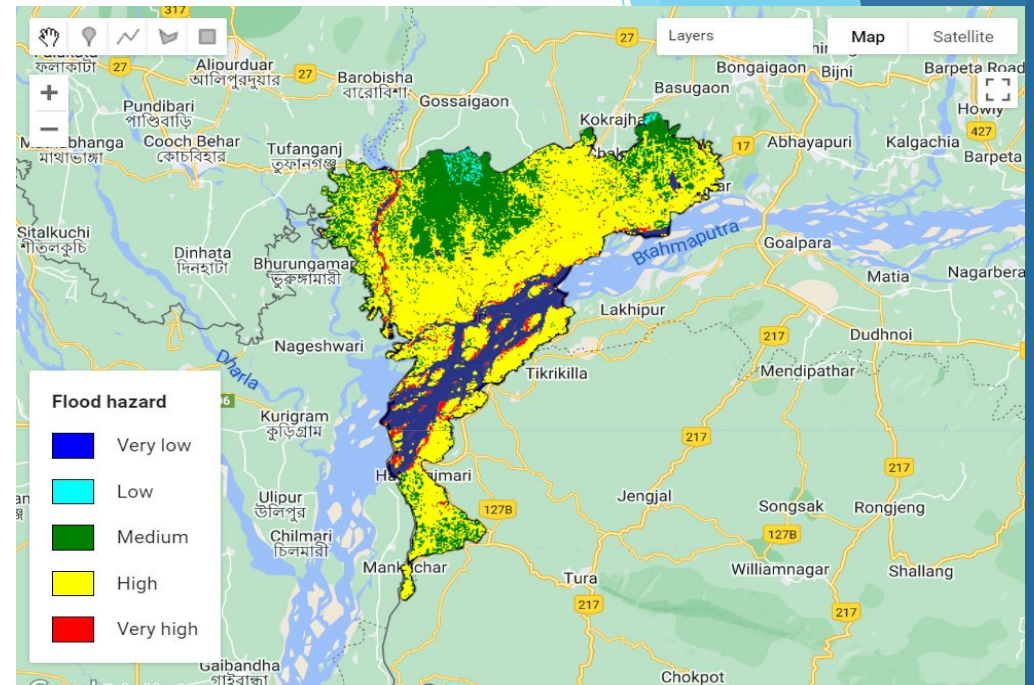
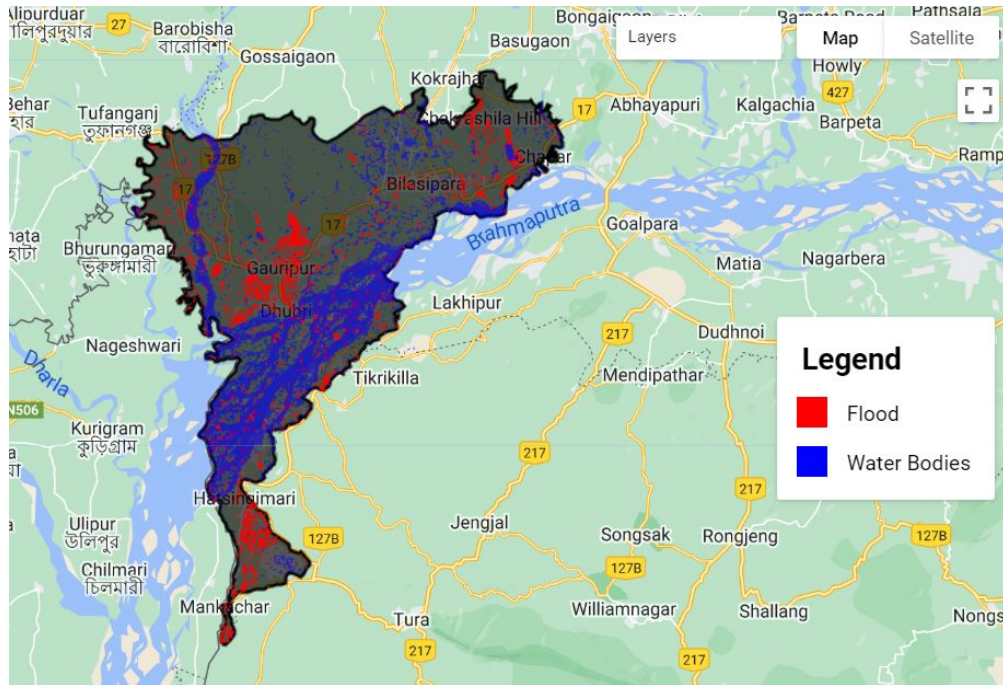
Tool for disaster prediction and assessment.

TEAM SAFER

@amn-d
@JanhaviSK
@VedantM21
@SonaSonarghare

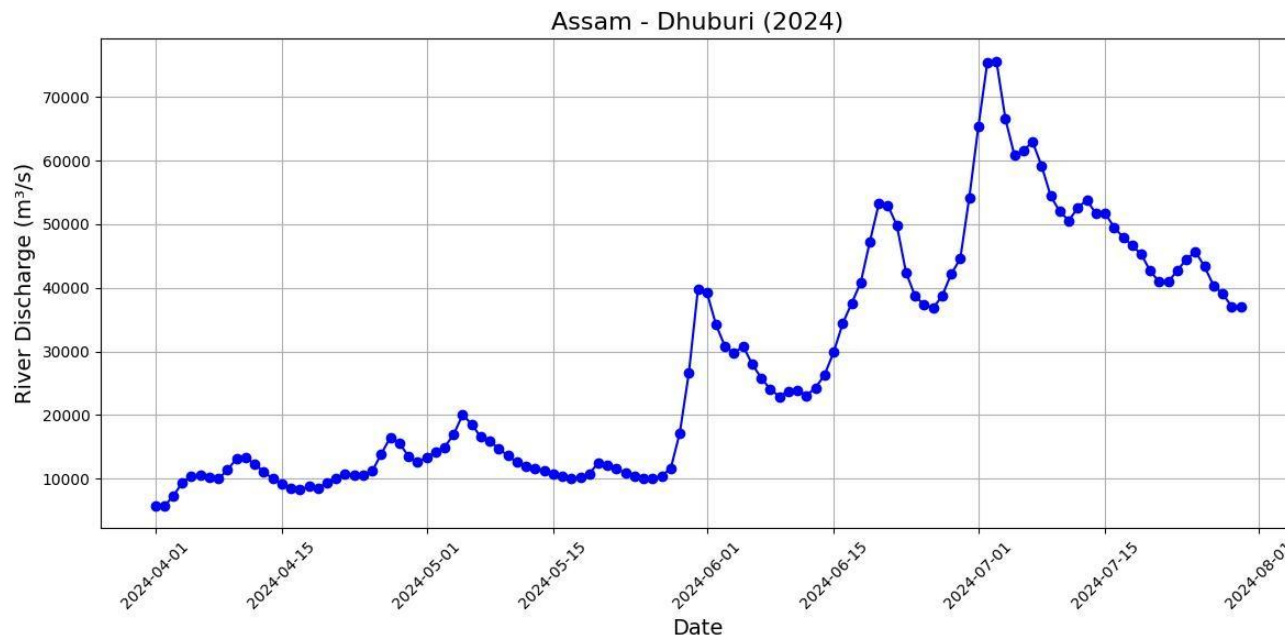
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Implementation Status



Flood Assessment

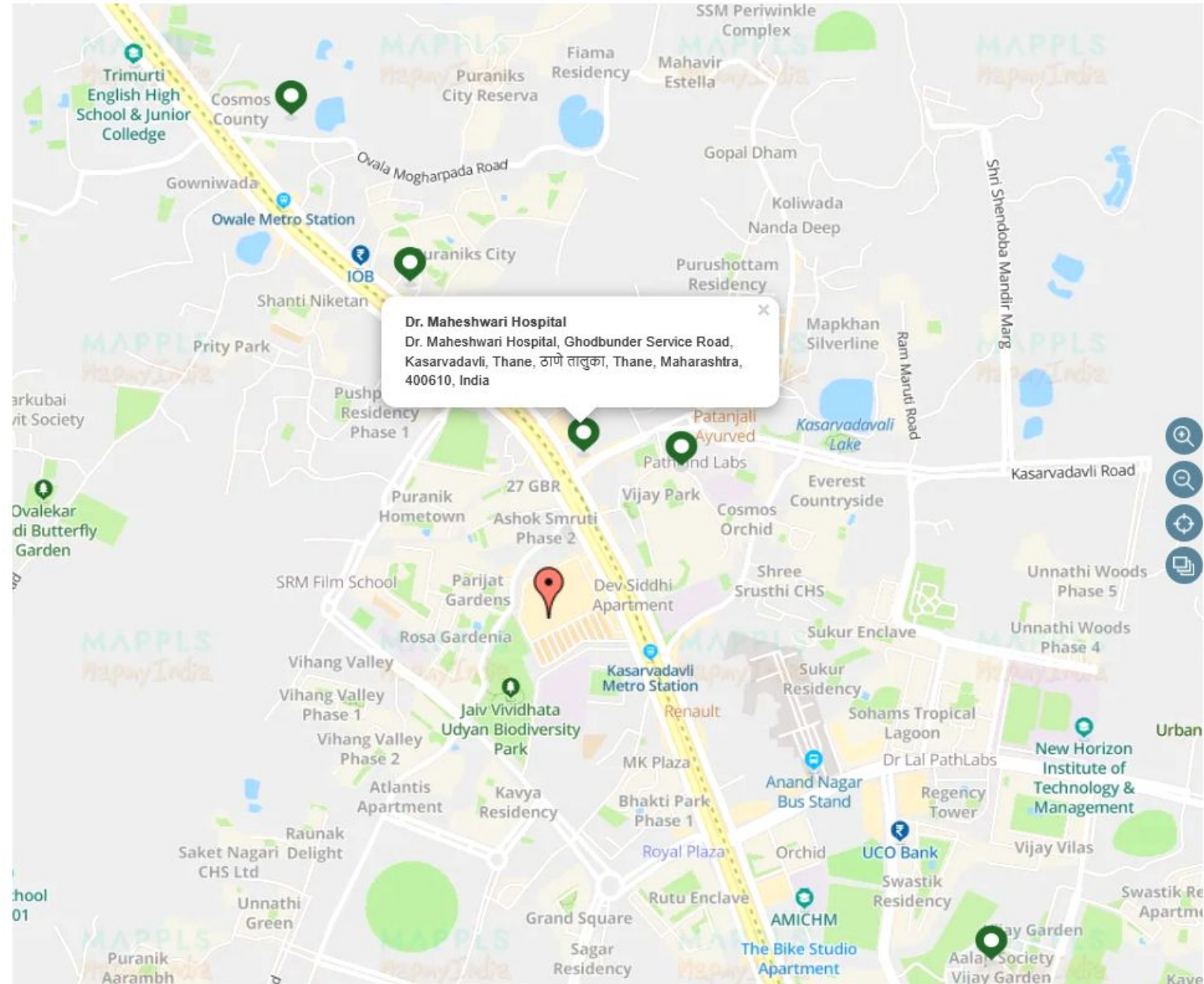
Flood Hazard Risk Map



Implementation Status

SOS

Showing nearest hospitals.



First Aid Now

By using this, you are agreeing to our terms and services. Team Safer will not be responsible for any damage that may occur. Observe the symptoms of your patient carefully, and proceed by selecting the 'Symptoms' button below.

Select Symptoms

Possible Diagnoses

Kindly select the symptoms from the options above to receive a diagnosis.

Fluid Buildup in Lungs

fluid buildup in the lungs, causing breathing difficulty.

Wound Infection

A cut or wound can become infected if bacteria enters.

Fracture

A fracture is a break in the bone, often caused by falls or heavy impact.

Symptoms of Pulmonary edema

Understanding Pulmonary edema

Pulmonary edema is fluid buildup in the lungs, causing breathing difficulty.

Treat Now

Symptoms

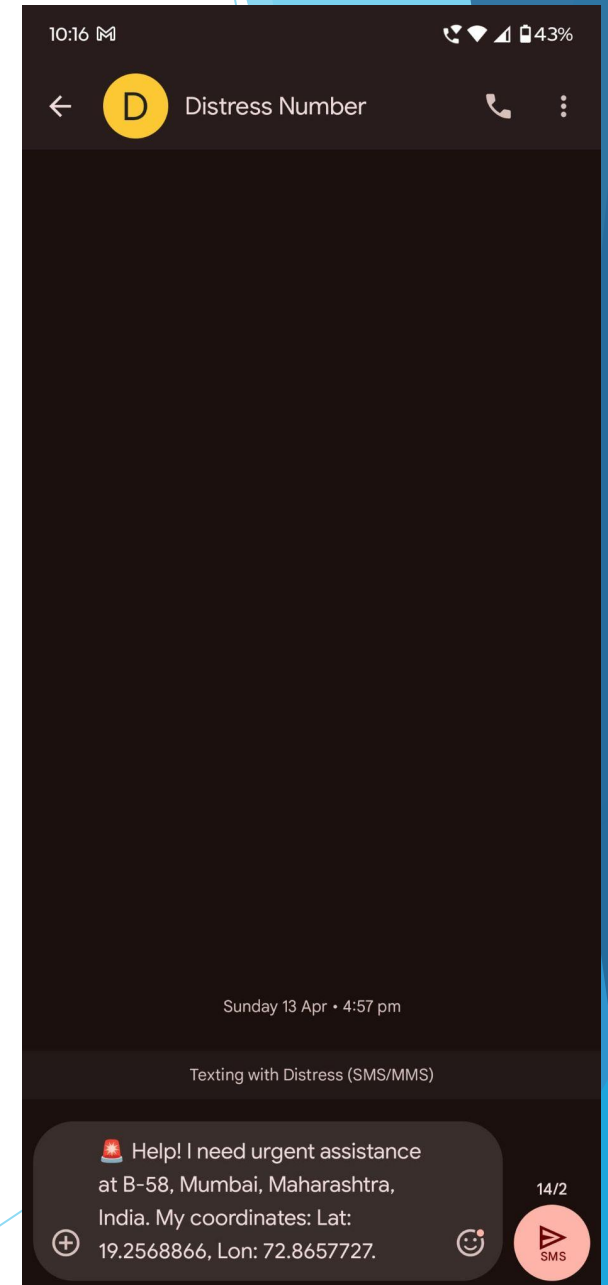
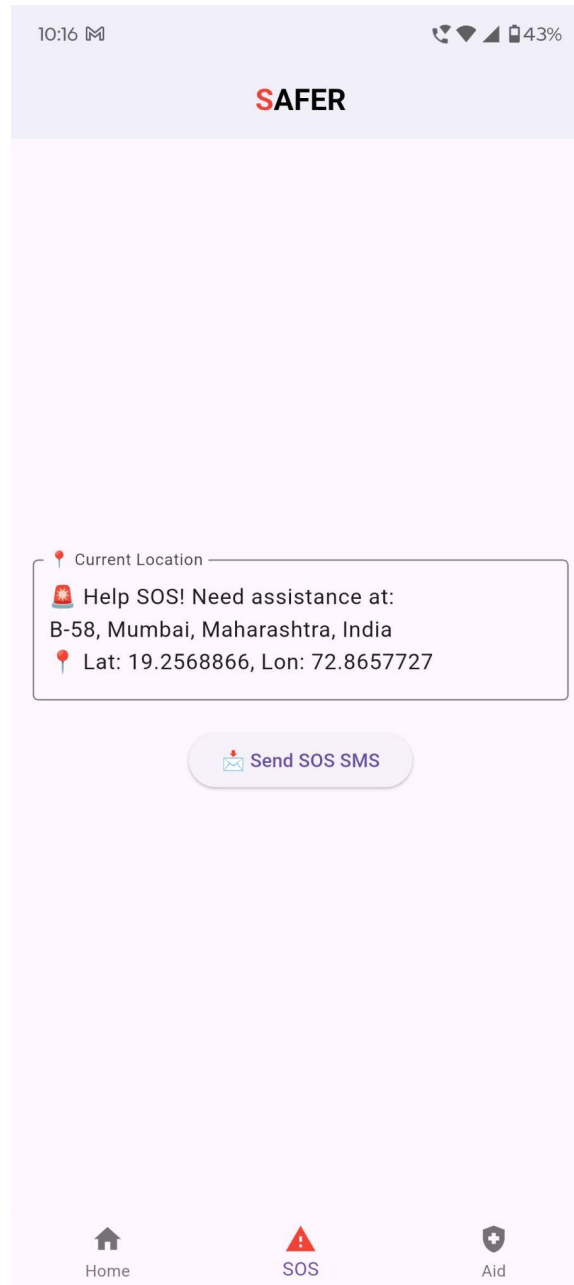
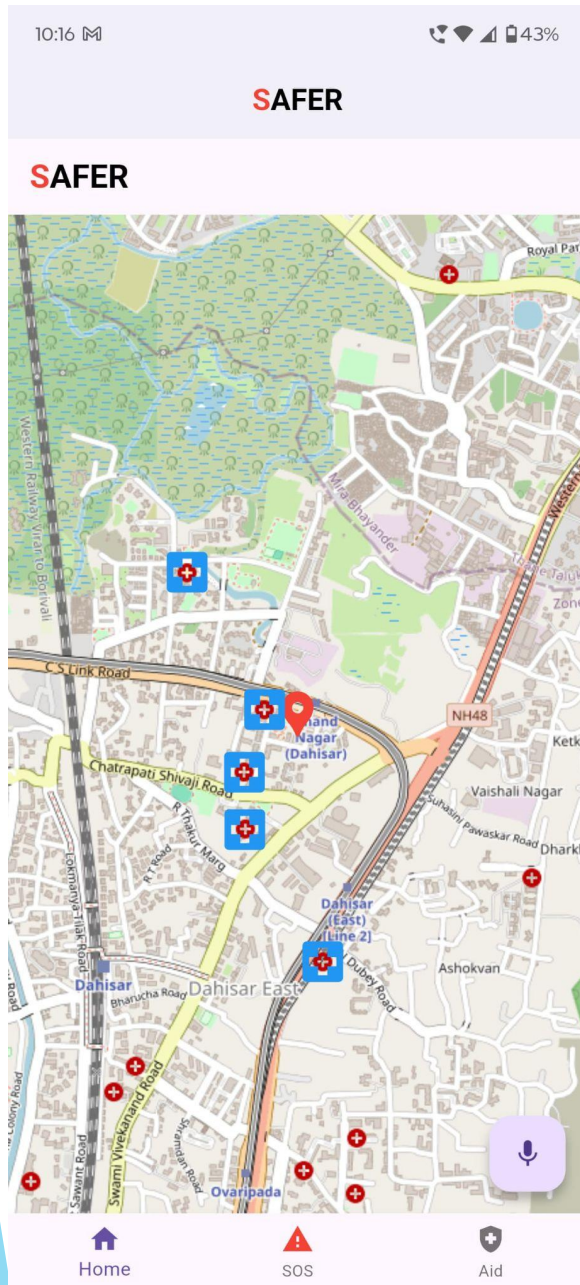


Tool for disaster prediction and assessment.
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Implementation Status



References

[1] Jinyang Du, John S. Kimball, Justin Sheffield, Ming Pan, Colby K. Fisher, Hylke E. Beck, Eric F. Wood, "Satellite Flood Inundation Assessment and Forecast Using SMAP and Landsat," IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, Volume 14, Pages 6707-6715, 2021.

<https://ieeexplore.ieee.org/document/9465655>

[2] S. Puttinaovarat, P. Horkaew, "Flood Forecasting System Based on Integrated Big and Crowdsourced Data Using ML Techniques," IEEE Access, Volume 8, Pages 5887-5893, 2020.

https://www.researchgate.net/publication/338376683_Flood_Forecasting_System_Based_on_Integrated_Big_and_Crowdsourced_Data_by_Using_Machine_Learning_Techniques

[3] Kuldeep Chaurasia, P. K. Garg, Rahul Dev Garg, "Geospatial Techniques for Flood Inundation Mapping," IEEE International Geoscience and Remote Sensing Symposium (IGARSS), July 2016.

https://www.researchgate.net/publication/309778410_Geospatial_techniques_for_flood_inundation_mapping

References

[4] Ghorbanzadeh, O., Shahabi, H., Crivellari, A. et al. Landslide detection using deep learning and object-based image analysis. *Landslides* 19, 929–939 (2022).

<https://link.springer.com/article/10.1007/s10346-021-01843-x>

[5] E. Collini, L. A. I. Palesi, P. Nesi, G. Pantaleo, N. Nocentini and A. Rosi, "Predicting and Understanding Landslide Events With Explainable AI," in *IEEE Access*, vol. 10, pp. 31175-31189, 2022.

<https://ieeexplore.ieee.org/document/9732490>

[6] Yang Hong, Robert F. Adler, George Huffman, "An Experimental Global Prediction System for Rainfall-Triggered Landslides Using Satellite Remote Sensing and Geospatial Datasets," *IEEE Transactions on Geoscience and Remote Sensing*, Volume 45, Issue 6, Pages 1671-1680, June 2007.

<https://ieeexplore.ieee.org/document/4215054>

References

[7] Praharaj, S., and E. Wentz. "Building community resilience through geospatial information dashboards." The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences 48 (2022): 151-157.

<https://isprs-archives.copernicus.org/articles/XLVIII-4-W5-2022/151/2022/>

[8] R. Keerthana, Dr. T. Ananth Kumar, P. Manjubala, M. Pavithra, "An Interactive Voice Assistant System for Guiding the Tourists in Historical Places," 2020 International Conference on Communication and Signal Processing (ICCSP), Pages 1-6, 2020.

<https://ieeexplore.ieee.org/abstract/document/9262347>

[9] Michael Raj T.F., Rajakumar B., Swaminathan S., Ramkumar M., "A Novel Approach: Voice Enabled Interface with Intelligent Voice Response System to Navigate Mobile Devices for Visually Challenged People," 2013 IEEE International Conference on Computer Communication and Informatics (ICCCI), Pages 1-6, 2013.

<https://ieeexplore.ieee.org/document/6496535>

References

- [10] Anbarasu P, Arunkumar S, Gowtham R, Deepak S, Latha M, "The Ambulance Booking and Tracking System," International Research Journal of Modernization in Engineering Technology and Science, Volume 06, Issue 04, Pages 1-10, April 2024.
https://www.irjmets.com/uploadedfiles/paper//issue 4 april 2024/54412/final/fin_irjmets1714503331.pdf
- [11] R. Jyothsingh, Thota Swathi, Kareti Lepakshi, Pulivarthi Akash, Puvvaladasu Hemanth, "Ambulance Booking System," International Journal of Innovative Science and Research Technology, ISSN No: 2456-2165, April 2024.
<https://doi.org/10.38124/ijisrt/IJISRT24APR2102>.
- [12] Dhyan Desai Dhyan, Joyeeta Dey, Sandeep Kumar Satapathy, Shruti Mishra, Sachi Nandan Mohanty, Pallavi Mishra, Sandeep Kumar Panda, "Optimal Ambulance Positioning for Road Accidents with Deep Embedded Clustering," IEEE Access, Volume 11, Pages 59917-59934, 2023.
<https://ieeexplore.ieee.org/document/10147832>

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